Dawood University Of Engineering and Technology, Karachi

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<u>BS ARTIFICIAL INTELLIGENCE</u>

Computer Vision Assignment Report

Pose Estimation & Activity Classification using MediaPipe Pose

1. Introduction

This project implements a complete pose estimation and activity classification pipeline using Computer Vision techniques. The system detects human body landmarks from video frames, computes important joint angles, and classifies activities such as sitting, standing, raising arms, and squatting using a rule-based classifier. The implementation uses the MediaPipe Pose model for human pose detection.

2. Video Source

The following YouTube video was used as the input source for activity detection and classification:

Video Link: https://youtu.be/ITv-_BkcrD0?si=tr1N1beYWCool6QP

3. Technologies and Libraries Used

Library	Purpose
OpenCV (cv2)	Video processing and frame handling
MediaPipe	Pose estimation and body landmark detection
NumPy	Mathematical computations and angle calculations
Matplotlib	Visualization and plotting graphs
SciPy	Signal smoothing and filtering

4. System Workflow

1. Input video is loaded using OpenCV.
2. Each frame is converted from BGR to RGB format.
3. MediaPipe Pose detects body landmarks.
4. Skeleton connections are drawn on the frame.
5. Joint angles are computed using geometric formulas.
6. Activities are classified using rule-based thresholds.
7. Results are visualized and saved as output video and graphs.

5. Joint Angle Computation

Joint angles are computed using three landmark points. The cosine rule is used to calculate the angle between vectors.

Formula Used:
Angle = $\arccos[(BA \cdot BC) / (|BA| \times |BC|)]$

The following joint angles were tracked in the project:

- Left Knee Angle
- Right Knee Angle
- Left Hip Angle
- Right Hip Angle
- Left Elbow Angle
- Right Elbow Angle
- Left Shoulder Angle
- Right Shoulder Angle

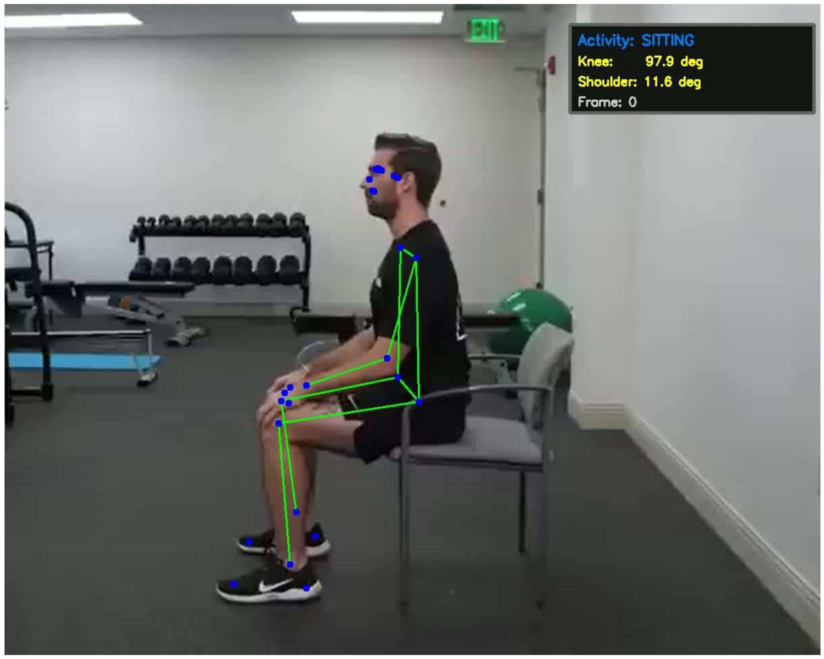
6. Rule-Based Activity Classification

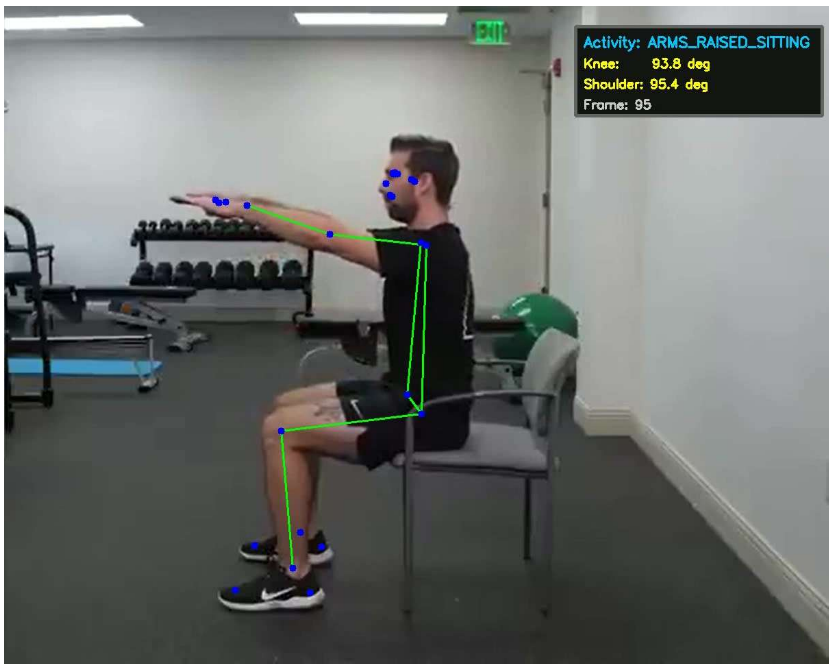
The activity classification system is based on threshold values of knee and shoulder angles.

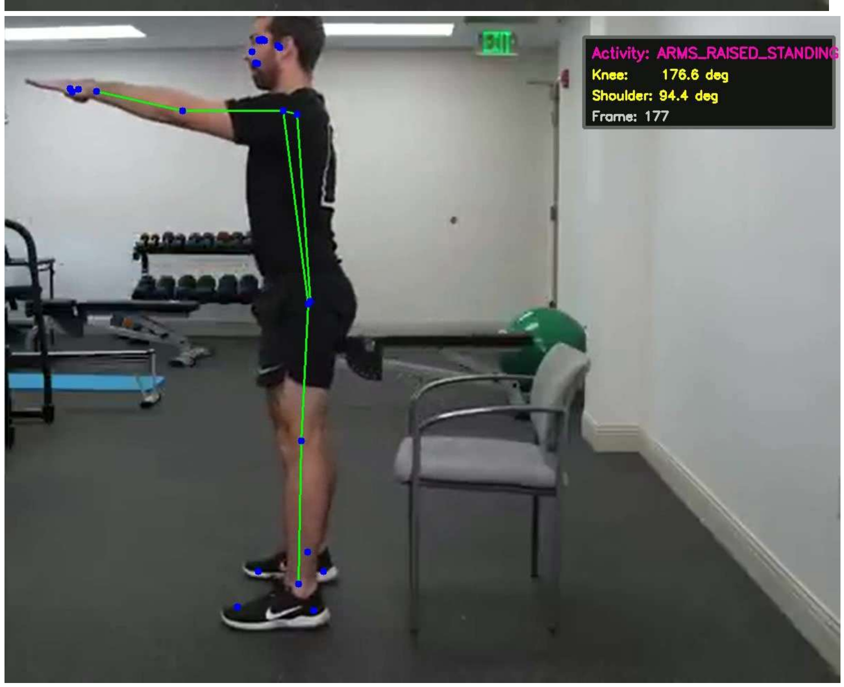
Activity	Condition	Description
SITTING	Knee Angle < 115°	Person is in sitting posture
ARMS_RAISED_SITTING	Knee < 115° and Shoulder ≥ 50°	Person sitting with raised arms
STANDING	Knee ≥ 165°	Person standing upright
ARMS_RAISED_STANDING	Knee ≥ 165° and Shoulder ≥ 50°	Standing with raised arms
SQUATTING	Intermediate knee angles	Detected during transition or squat posture

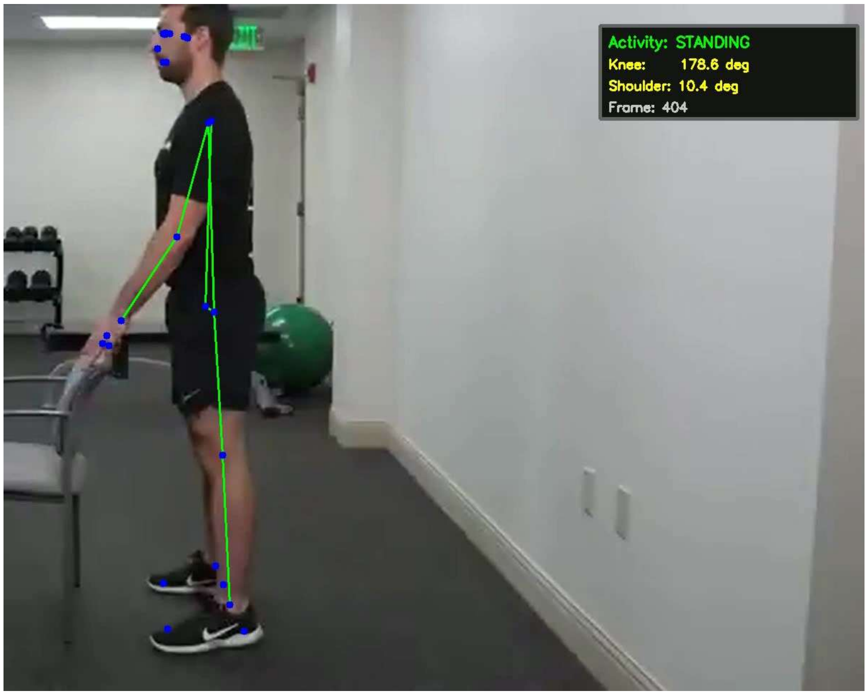
7. Results and Visualizations

The system successfully detected poses and classified multiple activities throughout the video sequence. Joint angle variations and activity transitions were visualized using graphs.


Activity: SITTING
Knee: 97.9 deg
Shoulder: 11.6 deg
Frame: 0


Activity: ARMS_RAISED_SITTING
Knee: 93.8 deg
Shoulder: 95.4 deg
Frame: 95


Activity: ARMS_RAISED_STANDING
Knee: 176.6 deg
Shoulder: 94.4 deg
Frame: 177


Activity: STANDING
Knee: 178.6 deg
Shoulder: 10.4 deg
Frame: 404

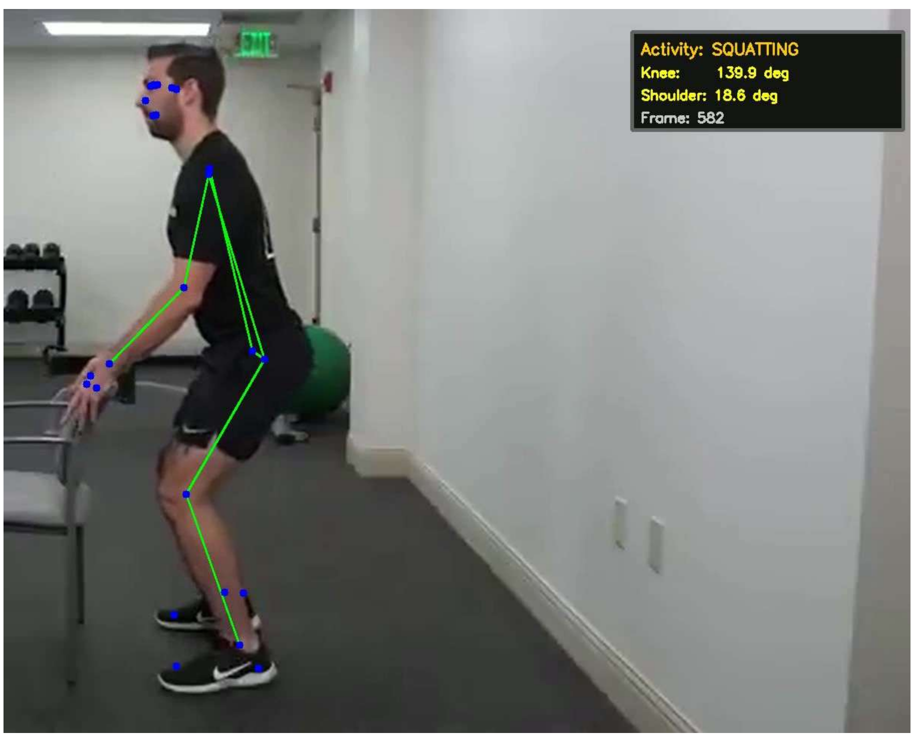

Activity: SQUATTING
Knee: 139.9 deg
Shoulder: 18.6 deg
Frame: 582

Figure 1: Joint Angle Tracking Graph

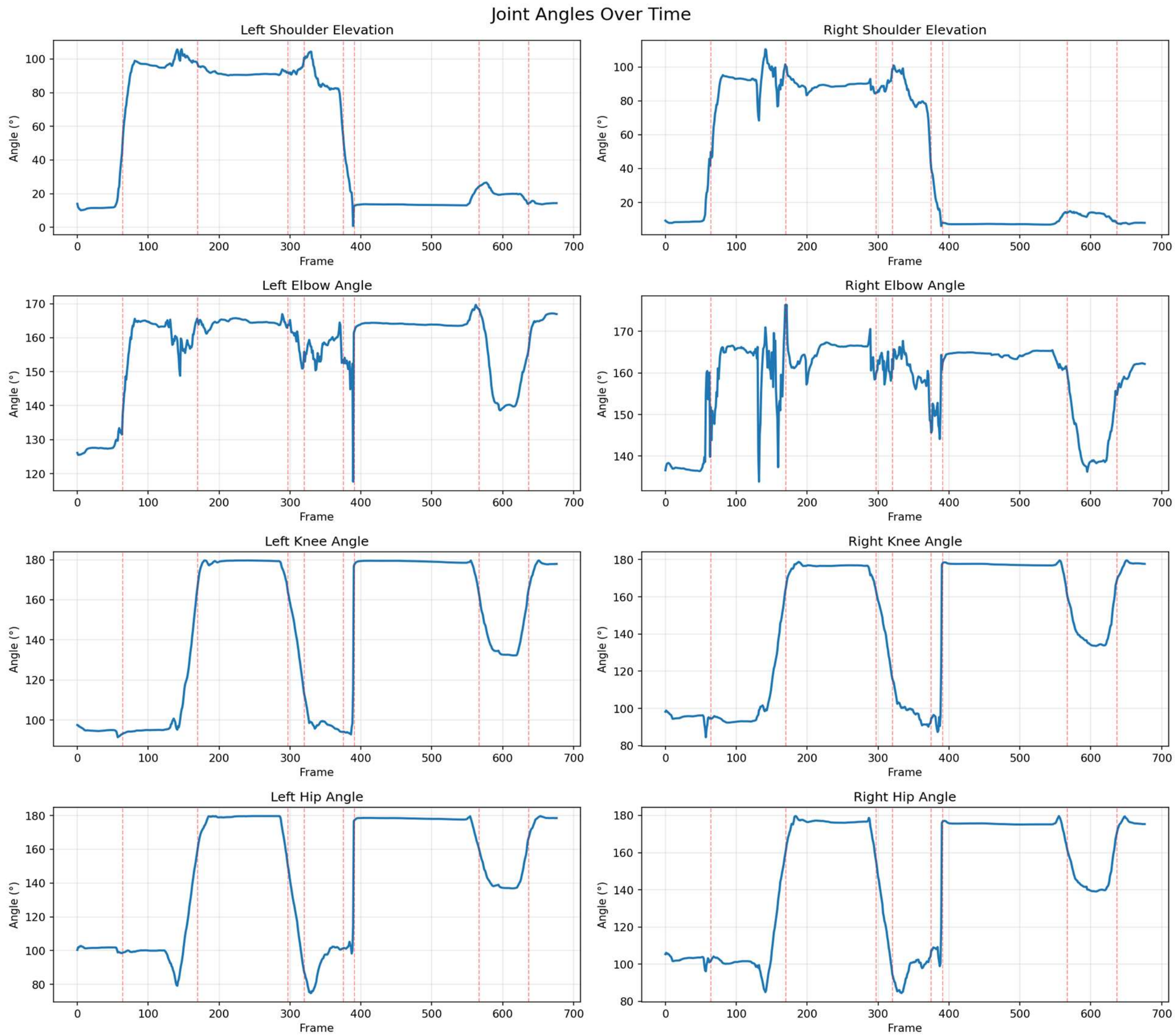
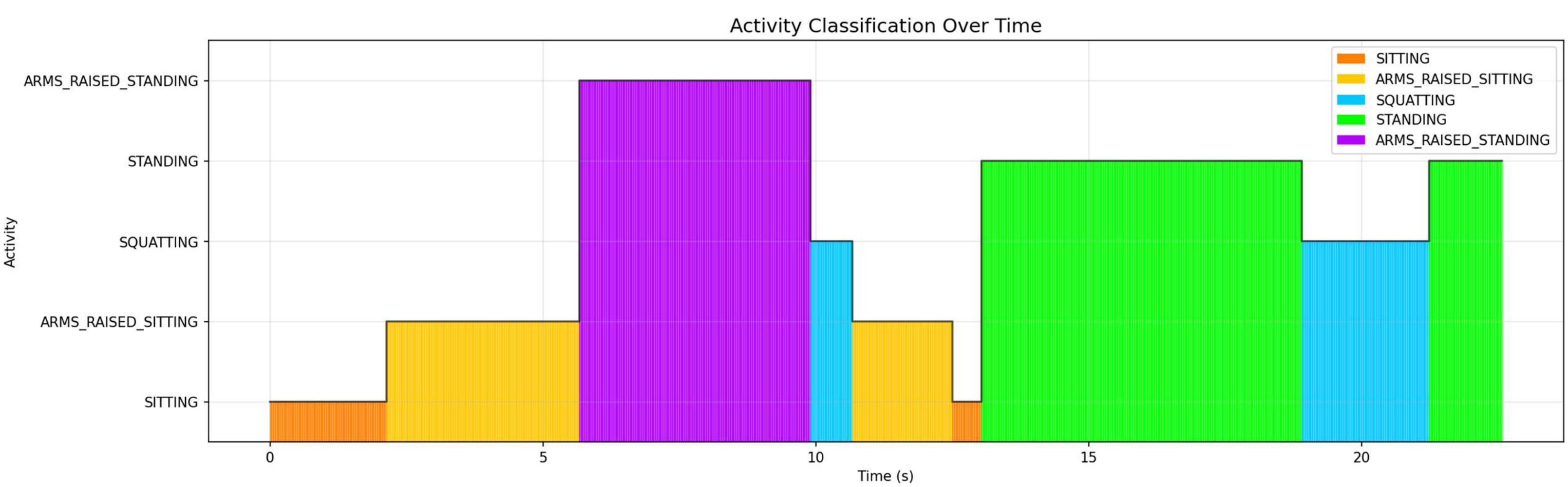


Figure 2: Activity Classification Timeline



8.Observations

- The MediaPipe Pose model successfully detected human body landmarks throughout most video frames.
- Skeleton tracking remained stable after applying smoothing techniques to reduce pose jitter.
- Joint angles changed significantly during activity transitions such as sitting to standing and arm raising movements.
- The rule-based classifier accurately identified activities including SITTING, STANDING, SQUATTING, and ARMS_RAISED_STANDING.
- The system processed all 678 video frames successfully at 30 FPS resolution.
- The overall activity classification accuracy achieved was **94.74%**, showing reliable performance of the designed rule-based system.
- Minor misclassifications occurred during rapid body movement and transition frames between activities.
- The generated angle plots and activity timeline clearly visualized posture changes across the video sequence.

9. Accuracy Evaluation

Manual ground truth labels were created for selected video frames containing different activities such as sitting, standing, squatting, and arm raising. Frames were selected from different parts of the video to ensure proper evaluation during posture transitions and stable movements. The predicted activity labels generated by the rule-based classifier were compared with the manually labeled ground truth frames to calculate the overall classification accuracy. The system achieved an overall accuracy of 94.74%.

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Classification Accuracy: 94.74%
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10. Conclusion

This project demonstrates the use of Computer Vision and Pose Estimation for real-time human activity recognition. By combining MediaPipe Pose, joint angle analysis, and rule-based classification, the system successfully identified multiple human activities from video data. The project also highlights the importance of preprocessing, smoothing, and geometric analysis in human pose understanding.